What about overcoming the Germ Theory?

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Research Overview

Science is not a static knowledge. Scientists are always looking for a new understanding of everything and society must apply this constant evolution in its own way to evaluate phenomena and react in abnormal situations.

One year passed since COVID-19 was announced but we're still facing this pandemic using a theory from XIX century, Germ theory, ignoring all the new findings achieved in the laboratories and this has a negative impact in the society.

Social relations, economy and freedom cannot be victims of an old theory that is not finding reason to still be used nowadays. Health is an inner chemical equilibrium and it cannot be dependent on solutions that don't integrate Biochemical new concepts and their mutual relations.

COVID-19 is showing us that we need to update our vision about Virus and how to prevent and cure them and this is my vision following recent researches.

Biochemical "New" concepts

Human Microbione [1] and Virome [2]

In 2012, Science mapped the entire Human Microbiome for the first time and found that more than 10.000 microbial species occupy the human ecosystem, corresponding to 81-99% of all microorganismal genera in healthy adults and this pletora contribute more genes responsible for human survival than humans contribute (360 times more bacterial genes). Bacterias can even replace themselves in order to maintain a specific metabolic function, so microbiome can change over time to achieve equilibrium in metabolic functions.

Everyone carries pathogens in equilibrium with its own microbiome and they cause no disease to healthy people but they may cause death to unhealthy people, so our efforts should be more focused in understanding this mechanisms in order to promote health for everyone and less focused in testing pathogens in everyone and fear their contagious behaviour.

A sepsis may occur in COVID-19. The news say that people are dying from hospital bacterias but these findings mean that people could be dying from inner bacterias while they are trying to maintain life metabolic functions, imbalanced in some conditions. A way to disrupt their intervention could be giving the right metabolic help to people.

HERVs [3]

Human Endogenous Retroviruses (HERVs) are ancient virus that incorporated human DNA million years ago and are still present. They represent 8% of total Human DNA. They suffered mutations along the way and contributed to Human evolution, some basic functions are nowadays result of HERVs as in the human placenta, but most of them have been silenced. In some conditions they can be activated and they are starting to be considered active in some diseases (Cancer, Diabetes, Alzheimer…) even if their action is just beginning to be understood.

Oxidative Stress [4]

Imbalance between free radicals (Reactive Oxygen Species) production and antioxidants, caused by age, diet, lifestyle, pollution, radiation and even body's natural immune response. Oxidative stress can accelerate the aging process and may contribute to the development of a range of chronic conditions including cancer, diabetes, heart disease and much more. All of this conditions are risk factors in COVID-19, proving that Oxidative Stress is involved in the severity of the disease.

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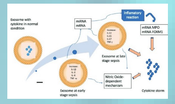
Hypothetical Mechanism for SARS-CoV-2 body creation

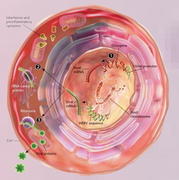
In XXI century, we cannot deal with a virus like it just can be an outside enemy to extremely avoid, fearing contagious, when we have so much virus and bacterias inside us contributing to our life and health and so much yet to learn about life and health natural processes.

Complete genome analysis and comparisons showed that COVID-19 did not originate from a known biological source or other previously characterized strains and HERVs may have a role in its range of symptoms [5]. The *orf1b* genetic sequence of SARS-CoV-2 has a similarity to the human gene NTNG1 [6]and the Netrin family may be a prognostic indicator for lung and kidney cancer [7].



Neutrophiles could prevent tumor metastasis via the generation of cytotoxic substances [8] and the incidence of cytokine storm is influenced by the presence of mRNA associated with inflammatory reactions such as MPO mRNA and FOXM1 mRNA. In addition, the nitric oxide dependent mechanism which is influenced by exosome [4] at early and late stage sepsis will modulate the occurrence of cytokine storm [9].

FOXM1 imbalanced is also related with smoke or aging and chronic diseases [10], so eldery and other risk factors people may present this oncogene ready to bind DNA and transcript related HERV sequence or other specific cell-cycle genes [11], such as NTNG1, following the usual HERV cycle production of viral proteins.



Credit: *https://www.the-scientist.com/infographics/infographic-65262*

Oncogene (1) Activation of viral promoters: Ancient retroviral infections have left viral promoters throughout the human genome. Although our bodies have coopted many of them to drive the expression of our own genes, a lot of those promoters are kept silenced through epigenetic repression. Reactivation of these elements can result in abnormal expression of nearby oncogenes or tumor-suppressor genes.

Viral promoter (2) Expression of viral genes: Under some circumstances, such as cancer, many regions of the genome that are normally silenced can awaken. This can activate the transcription of HERVs, causing viral RNA to accumulate in the cytoplasm. According to the "viral mimicry" theory, these molecules alert cellular RNA sensing pathways to the viral material, triggering an immune response.

Viral proteins (3) Translation of viral proteins: Some viral RNAs are translated into proteins, which can be secreted and travel to other cells. It's unclear what effects these proteins have, but some researchers hypothesize they activate surface receptors and ultimately initiate immune reactions

A research to explain SARS-CoV-2 reinfections [12] says that a post-mortem histopathological analysis showed intracellular viruses organized into arrays, indicating that they were manufactured and assembled intracellularly. These findings also suggested that there is a possibility for the use of the exosomal cellular transport as a mode of systemic SARS-CoV-2 viral dissemination and serve as a potential means for the COVID-19 reactivation.

What if the reactivation means that this cycle just didn't stop yet and the viruses were manufactured intracellularly since the first one, as a way to protect from Cancer or as an imbalance originated by the other chronic diseases or aging (risk factors, all dependent on Oxidative Stress)? What if that's why a COVID-19 cure also cured a Cancer [13]?

Natural Solutions

Quercetin and Vitamin C [4]

Quercetin (a natural flavonoid mainly found in red fruits and vegetables) and Vitamin C (ascorbic acid mainly found in citrics fruits) had a lot of previous research in several diseases. It has been proved that they have antiviral, anticancer, immunomodulatory and antioxidant effects, and Quercetin antiviral effect is enhanced by Vitamin C. Vitamin C also proved that sometimes disease is not caused by germs, but instead by nutritional deficits (Scurvy) even if everyone is infected and it looks like contagious.

Quercetin and Vitamin C may act inhibiting virus entry in the cell, virus replication, enzyme activity and assembly; fortifying immunity (modulating interleukins, promoting early INFS production, T cell maturations and fagocity activity) and fighting free radicals, reducing Oxidative Stress in the body. Comparing all this effects with vaccines, we can say that Quercetin and Vitamin C is a more complete prevention with no side effects in normal consume as food. There are some clinical trials ongoing to know if there's any side effects in using supplements to prevent SARS-CoV-2 but they usually haven't known side effects till 1g daily dosage). They are also low cost and widely available.

Quercetin and Vitamin C is a natural solution recommended in scientific literature for profylaxis and cure of mild and severe cases of COVID-19 [14] and this should be notice in the news instead of all kind of limitations, because of the effective action against the virus, the enhancing of immunity and the risk factors prevention, including cancer.

	Quercetin	Vitamin C
Prophylaxis	250–500 mg BID	500 mg BID
Mild cases	250–500 mg BID	500 mg BID
Severe Cases*	500 mg BID	3 gr q6 for 7 days

**ARDS-like presentation, require assisted ventilation/intubation, ICU hospitalization.*

Glutathione [4]

Glutathione is an endogenous antioxidant present in all cells (except neurons). COVID-19 critical cases and deaths are related with a low GSH/GSSH ratio which indicates high levels of oxidative stress. N-acetylcystein (NAC) is a Glutathione precursor and promote a fast recuperation of this ratio. It's important to reinforce all natural chemicals related to Glutathione metabolic cycles, in order to avoid a cytokines storm: Vitamin D and Magnesium, Vitamin B12, Melatonin, Zinc and Selenium.

Rethinking Global Prevention

PCR test

PCR is an amplification of genetic material, it is not an effective way to diagnose disease. There are few primers used in PCR COVID-19 test and there's too much possible source of unknown RNA in the body (exosomes) [4] that can make it be not so specific as thinking and cause false positives and false alarm in the society.

Social Distancing

Virus doesn't determine how body react, the same SARS-CoV-2 virus can cause no symptoms, death and all in between, so it is the inner state of each person that determines its effect. The best way to prevent COVID-19 is to reinforce immune system, healthy lifestyle and antioxidants, instead of isolation and fear that virus can be extremely contagious and dangerous by itself, from the outside, in all conditions.

Mask

SARS-CoV-2 may enter cells via acidic endosomes [4], so it's crucial to maintain endosomes with the highest pH possible to minimize this attack. Blood uses to be slightly alkaline (pH=7.4) and endosomes are usually already acidic. Alkaline water may help to maintain us more protected [15]. Masks force to breathe CO2 and this can decrease blood pH and endosomes pH and even a little variation can enhance endosome acidification. Masks don't even filter virus in absolute, so instead of prevention, masks can promote virus attack and infection, while dramatically increase pollution in our oceans without preventing COVID-19.

Vaccines

Vaccines are still in testing phase so they should not be massively used and presented as perfect solutions until the end of all testing phases. People are suffering side effects [16] (including death in early age without previous health complications) after vaccination and this should be considered enough to rethink these solutions even if using the same old Germ theory. Vaccines only protect from one virus protein, maybe that's why people continue to be infected after vaccination. The natural response to the natural virus would be a better complete vaccine and it is not dangerous if we protect ourselves with the natural solutions referenced.

Conclusions

A XXI century theory must emerge: Virus and Bacterias are natural in Human beings and they only manifest and provoque disease in imbalanced conditions, while trying to maintain metabolic functions. Oxidative Stress is the initiator of this imbalance and the cause of death, so Antioxidants are a most effective solution to fight COVID-19 and its risk factors: Flavonoids rich diet in a sinergistic action with Alkaline water, Vitamins B12, C and D, Magnesium, Zinc and Selenium, good sleep, no smoke and better environmental conditions. NAC can be a solution to critical cases.

References

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Recommended searches: Quercetin, Vitamin C, Vitamin D, Vitamin B12, Magnesium, Melatonin, Glutathione, Selenium, Zinc, NAC (N-acetylcystein), Serotonin, Luteolin, Exosome, Microbiome, Virome, PM2.5, Oxidative Stress, Hervs

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